

Differential Equations and Vector Calculus

SITAMS CSE (AI & ML) • R23 • I Year • II Semester • Course Code: 23BSC121T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: DIFFERENTIAL EQUATIONS OF FIRST ORDER AND FIRST DEGREE

Syllabus topics

- Linear differential equations – Bernoulli's equations - Exact equations and equations reducible to exact form
- Applications: Newton's Law of cooling – Law of natural growth and decay, Electrical circuits

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit II: LINEAR DIFFERENTIAL EQUATIONS OF HIGHER ORDER (CONSTANT COEFFICIENTS)

Syllabus topics

- Definitions, homogenous and non-homogenous, complimentary function, general solution, particular integral, Wronskian, Method of variation of parameters
- Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion

Revision checklist

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Unit III: PARTIAL DIFFERENTIAL EQUATIONS

Syllabus topics

- Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method and Non-Linear (Standard forms) equations
- Homogeneous Linear Partial differential equations with constant coefficients (Method of Separation of variables)

Revision checklist

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Unit IV: VECTOR DIFFERENTIATION

Syllabus topics

- Scalar and vector point functions, vector operator Del, Del applies to scalar point functions Gradient, Directional derivative, del applied to vector point functions-Divergence and Curl, vector identities

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Unit V: VECTOR INTEGRATION

Syllabus topics

- Line Integral-circulation-work done, surface integral-flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and related problems

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